Derivate

Sia $f: X \subseteq \mathbb{R} \to \mathbb{R}$, f è derivabile in $x_0 \in X$

se $\exists$ finito

$$\lim_{x \to x_0} \underbrace{\frac{f(x) - f(x_0)}{x - x_0}}_{\text{RAPPORTO INCREMENTALE}} = f'(x_0)$$

$P(x, x_0)$

↳ DERIVATA DI f in x_0

la retta $y = f(x_0) + m(x - x_0)$ → RETTA TANGENTE A f in $(x_0, f(x_0))$

(m → coefficiente angolare)

è la migliore approssimazione lineare di f in x_0.

$$f(x) = f(x_0) + m(x - x_0) + o(x - x_0) \quad \text{per } x \to x_0$$

$T_1(x)$

$$\lim_{x \to x_0} \frac{f(x) - f(x_0) - m(x - x_0)}{x - x_0} = 0$$

+ mi avvicino a x_0
+ sbaglio meno

$$E_1(x) = f(x) - T_1(x)$$ è infinitesimo per $x \to x_0$

NOTA:

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

$P(x_0+h, x_0)$

Derivate elementari

0) $f(x) = K$ $\qquad f'(x) = 0$

i) $f(x) = x^{\alpha}$ $\qquad f'(x) = \alpha x^{\alpha-1}$ $\qquad \forall x > 0$, $\forall \alpha \in \mathbb{R}$

DIM.

$$\lim_{h\to 0} P(x+h,h) = \lim_{h\to 0} \frac{\overbrace{(x+h)^{\alpha}}^{f(x+h)} - \overbrace{x^{\alpha}}^{f(x)}}{h} =$$

$$= \lim_{h\to 0} \frac{\left[x\left(1+\frac{h}{x}\right)\right]^{\alpha} - x^{\alpha}}{h} =$$

$$= \lim_{h \to 0} \frac{x^\alpha \left(1 + \frac{h}{x}\right)^\alpha - x^\alpha}{h} =$$

$$= \lim_{h \to 0} \frac{x^\alpha \left[\left(1 + \frac{h}{x}\right)^\alpha - 1 \right]}{h} \qquad \left(1 + \frac{h}{x}\right) \to 0$$

$$* \sim \frac{x^\alpha \cdot \alpha \cdot \frac{h}{x}}{h} = \alpha x^{\alpha - 1}$$

$$(1 + a_n)^\alpha - 1 \sim \alpha a_n$$

ii) $f(x) = e^x \qquad f'(x) = e^x \qquad \forall x \in \mathbb{R}$

Dim.

$$\lim_{h\to 0} \frac{e^{x+h} - e^x}{h} = \lim_{h\to 0} \frac{e^x e^h - e^x}{h} =$$

$$= \lim_{h\to 0} \frac{e^x \left(e^h - 1\right)}{h} \sim \frac{e^x \cdot h}{h} = e^x$$

* $e^{an} - 1 \sim an$

iii) $f(x) = a^x$ $\quad f'(x) = a^x \log a$ $\quad \forall x \in \mathbb{R}$, $\forall a > 0$

iv) $f(x) = \log |x|$ $\quad f'(x) = \dfrac{1}{x}$ $\quad \forall x \neq 0$

v) $f(x) = \log_a |x|$ $\quad f'(x) = \dfrac{\log_a e}{x} = \dfrac{1}{x \log a}$ $\quad \forall a > 0$, $a \neq 1$

vi) $f(x) = \sin x$ $\quad f'(x) = \cos x$ $\quad \forall x \in \mathbb{R}$

vii) $f(x) = \cos x$ $\quad f'(x) = -\sin x$ $\quad \forall x \in \mathbb{R}$

viii) $f(x) = |x| = \begin{cases} -x & \text{se } x < 0 \\ x & \text{se } x \geq 0 \end{cases}$ $\quad f'(x) = \begin{cases} -1 & \text{se } x < 0 \\ +1 & \text{se } x > 0 \end{cases}$ $\quad \forall x$

*

Osservazione parità/disparità e derivata prima

* f PARI $\Rightarrow$ f' DISPARI

  f DISPARI $\Rightarrow$ f' PARI

* DIM

$$f'(-x) = \lim_{h\to 0} \frac{f(-x+h) - f(-x)}{h}$$

$f(-x) = f(x)$ ✓

$$= \lim_{h\to 0} \frac{f(x-h) - f(x)}{h} \overset{\substack{k=-h \\ h}}{=} \lim_{k\to 0} \frac{f(x+k) - f(x}{-k}$$

$$\frac{}{x)} = -\lim_{k\to 0}\frac{f(x+k)-f(x)}{k} = -f'(x)$$

C.V.D.

$$f'(-x) = -f'(x)$$

Algebra delle derivate

Siano $f, g : X \subseteq \mathbb{R} \to \mathbb{R}$ derivabili in $x_0 \in X$.

Sono derivabili in x_0 anche $\alpha f + \beta f$, fg , $\frac{f}{g}$

con $\alpha, \beta \in \mathbb{R}$ e $g(x_0) \neq 0$.

SOMMA

i) $(\alpha f + \beta g)'(x_0) = \alpha f'(x_0) + \beta g'(x_0)$

PRODOTTO

ii) $(fg)'(x_0) = f'(x_0) \cdot g(x_0) + g'(x_0) \cdot f(x_0)$

iii) $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - g'(x_0)f(x_0)}{g(x_0)^2}$ **RAPPORTO** per $g(x_0) \neq 0$

Dim.

$$\left(\frac{1}{g}\right)'(x_0) = \lim_{x \to x_0} \frac{\frac{1}{g(x)} - \frac{1}{g(x_0)}}{x - x_0} =$$

$$= \lim_{x \to x_0} \frac{\frac{g(x_0) - g(x)}{g(x)\,g(x_0)}}{x - x_0} = \lim_{x \to x_0} \frac{g(x_0) - g(x)}{x - x_0} \cdot \frac{1}{g(x)g(x_0)} = -\frac{g'(x_0)}{g^2(x_0)}$$

($\lim_{x \to x_0} \frac{g(x_0) - g(x)}{x - x_0} = -g'(x_0)$)

$$\left(\frac{f}{g}\right)'(x_0) = \left(f \cdot \frac{1}{g}\right)'(x_0) =$$

$$= f'(x_0)\,\frac{1}{g(x_0)} + f(x_0)\cdot\left(-\frac{g'(x_0)}{g^2(x_0)}\right)$$

$$= \frac{f'(x_0)\cdot g(x_0) - f(x_0)\,g'(x_0)}{g^2(x_0)}$$

✳ (ix) $f(x) = tg\,x$ $\qquad f'(x) = \frac{1}{\cos^2 x} = 1 + tg^2 x$

(x) $f(x) = cotg\,x$ $\qquad f'(x) = -\frac{1}{\sin^2 x} = -1 - cotg^2 x$

Dim ix

$f(x) = tg\,x = \frac{\sin x}{\cos x}$ (sin → f, cos → g)

$f'(x) = \frac{\cos x \cdot \cos x - (-\sin x)\sin x}{\cos^2(x)} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$

Esempio

$$f(x) = 3x^2 - x\sin x$$

→ $(fg)'$ $\quad f(x) = x$, $g(x) = \sin x$

$$f'(x) = 3 \cdot 2 \cdot x^{2-1} - \left[1 \cdot \sin x + x \cdot \cos x\right] =$$

$$= 6x - \sin x - x\cos x =$$

$$= x\left(6 - \cos x\right) - \sin x$$

Esempio

$$f(x) = \frac{x^2 - 1}{3x - 4}$$

$$\left(\frac{f}{g}\right)' \qquad f(x) = x^2 - 1 \qquad g(x) = 3x - 4 \qquad f'(x) = 2x \qquad g'(x) = 3$$

$$f'(x) = \frac{2x(3x-4) - 3(x^2-1)}{(3x-4)^2} =$$

$$= \frac{6x^2 - 8x - 3x^2 + 3}{9x^2 - 24x + 16} = \frac{3x^2 - 8x + 3}{9x^2 - 24x + 16}$$

Esercizio ASINTOTI

$$f(x) = \frac{\log\left(e^{3x^2} - 2\right)}{x + 7}$$

DOMINIO

$$x + 7 \neq 0 \implies x \neq -7$$

$$e^{3x^2} - 2 > 0 \iff e^{3x^2} > 2 \iff 3x^2 > \log 2$$

$$\iff x^2 > \frac{\log 2}{3} \iff x < -\sqrt{\frac{\log 2}{3}} \quad x > +\sqrt{\frac{\log 2}{3}}$$

$$D = \left(-\infty, -\sqrt{\frac{\log^2}{3}}\right) \cup \left(\sqrt{\frac{\log^2}{3}}, +\infty\right) \setminus \{-7\}$$

$$\lim_{x \to -7^{\pm}} \frac{\log\left(e^{3x^2} - 2\right)}{x+7} = \frac{\log\left(e^{147} - 2\right)}{0^{\pm}} = \pm\infty$$

$$\boxed{x = -7}$$

$$\lim_{x \to \pm\sqrt{\frac{\log^2}{3}}} \frac{\log\left(e^{3x^2} - 2\right)}{x+7} = \frac{-\infty}{\pm\sqrt{\frac{\log^2}{3}} + 7} = \mp\infty$$

$$\boxed{x = \pm\sqrt{\frac{\log^2}{3}}}$$

AS. ORIZZ.

$$\lim_{x \to \pm\infty} f(x) = +\infty$$

AS. OBLIQUO

$$\lim_{x \to \pm\infty} \frac{f(x)}{x} = m$$

$$\lim_{x \to \pm\infty} f(x) - mx = q$$

$$y = mx + q$$

SOL.

$y = 3x - 21$